Bruno Lopez

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SKILLS

Data & Analytics: Python, R, SQL, Pandas, NumPy, Pytorch, Statistical Analysis, Data Cleaning, Time Series, Data Visualization, Power BI (DAX), Excel Automation
AI / ML: LLMs (Claude API, GPT, Gemini, Grok), BART, Longformer, Causal Forests, Random Forest, XGBoost, Mask R-CNN, NLP, Computer Vision
Geospatial: Google Earth Engine, Sentinel-1/2, Hyperspectral (AVIRIS-NG, Isofit), MapLibre GL, geemap, GIS, Ground Truthing
Infrastructure: Git/GitHub, Linux, Bash, Anaconda, GitHub Copilot, Copilot Studio, Power Automate, FastAPI, PostgreSQL, Streamlit, REST APIs, AWS S3

EXPERIENCE

Portfolio Management Analyst · **Pacific Gas & Electric Company (PG&E)** *Oakland, CA | Dec 2024 – Apr 2026*

- Architected and deployed a production LLM pipeline (Anthropic Claude API) to autonomously parse and extract structured data from broker emails. This eliminated a manual process worth several weeks of analyst labor annually.
- Drove department-wide AI adoption by introducing GitHub Copilot to a team with zero prior AI tooling, automating multiple projects and shifting default development workflows across the org.
- Managed the department's SQL Server database: designed and maintained tables and views, integrated multi-source energy data, and automated ingestion pipelines using Python and Power Automate.
- Built and maintained Python and SQL scripts to pull position data, broker quotes, and energy market information; managed the team's GitHub repository and enforced version control standards.
- Developed Power BI dashboards using DAX across multiple datasets, and automated legacy Excel workflows in Python. This Improved reporting speed and reduced manual overhead across teams.
- Served as one of two analysts statewide ensuring California meets CPUC/CAISO reserve capacity requirements under PG&E's Resource Adequacy program.

Energy Analyst · **Tesla** *Fremont, CA | May 2023 – Apr 2024*

- Deployed Transformer-based summarization models (BART, Longformer) to condense complex case notes across commercial solar sites, improving data accessibility and reducing time-to-insight for operational teams.
- Built ML algorithms to detect equipment failures across commercial solar PV systems (no power events, peak output drops, bad meter connections), preventing losses valued at several million dollars.
- Built Streamlit dashboards to visualize PV issues, contract renewals, and performance trends; engineered Python tools that transitioned multiple teams off Excel workflows.
- Mentored 2 interns in energy data science and led team-wide training on AI/ML workflows and GitHub best practices, improving team-wide analytical capability.

Remote Sensing Data Scientist · **Orbital Sidekick** *San Francisco, CA | Jan 2022 – Dec 2022*

- Built end-to-end ETL pipelines for hyperspectral satellite imagery in Python, from raw data ingestion through preprocessing, feature extraction, and model-ready outputs.
- Developed atmospheric correction models using NASA's Isofit framework, improving spectral fidelity of hyperspectral data for downstream ML tasks.
- Applied ML across multiple domains: crop speciation, fire change detection, time-series vegetation analysis, mineral mapping, and the NASA Harvest Project.
- Conducted field validation campaigns to ground-truth algorithm outputs, adjusting models based on real-world observations. I also operated the hyperspectral camera aboard a plane on occasion.
- Investigated computer vision approaches for hyperspectral imagery classification; developed reusable OOP Python modules to increase pipeline efficiency.

Research Data Analyst · **Stanford University** *Stanford, CA | Jan 2021 – Jan 2022*

- Co-authored a published paper applying causal forests to measure the real-world effectiveness of cover crops across the U.S., combining causal inference with large-scale agricultural data.
- Contributed to ML projects spanning crop field identification in Africa, object detection of rural structures, and soil characteristic analysis from web-scraped datasets.
- Applied satellite imagery to a human trafficking research project in Brazil, and assisted a Remote Sensing Computer Vision course. This supports both students and active research simultaneously.
- Automated manual Excel-based research workflows into R and Python pipelines, significantly accelerating the team's data processing capacity.

EDUCATION

B.S. Earth Science, Data Analytics Concentration (W/ Honors) · UC Santa Cruz *Graduated June 2020*